

BUGULMA, Russian Federation

WMO: 287110

Lat: 54.6329N

Lon: 52.8004E

Elev: 997

StdP: 14.17

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.5	-11.2	-23.0	1.6	-16.9	-16.8	2.3	-10.0	29.3	20.9	26.6	20.8	5.9	300	0.632

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.6	86.6	66.4	83.2	65.4	80.2	63.7	69.1	81.3	67.5	79.6	65.9	77.0	8.3	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.8	95.4	74.7	63.3	90.3	73.1	61.7	85.1	71.4	33.8	81.1	32.5	80.0	31.2	77.2	75.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
24.7	21.0	18.7	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-21.9	90.6	7.1	3.5	-27.0	93.1	-31.1	95.1	-35.1	97.1	-40.2	99.6		
			-22.1	71.5	7.0	2.2	-27.2	73.0	-31.3	74.3	-35.2	75.5	-40.3	77.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	39.1	11.0	11.9	23.1	41.1	55.0	63.0	66.3	63.5	52.9	39.7	25.0	14.4	(6)
(7)		DBStd	22.48	11.57	11.63	8.29	11.02	9.60	7.98	7.10	7.96	8.63	8.87	10.34	11.87	(7)
(8)		HDD50	5747	1209	1066	835	312	60	5	0	2	65	341	749	1103	(8)
(9)		HDD65	9764	1674	1486	1300	717	332	130	72	126	374	786	1199	1568	(9)
(10)		CDD50	1756	0	0	0	46	215	397	506	420	152	20	0	0	(10)
(11)		CDD65	297	0	0	0	1	22	71	113	79	11	0	0	0	(11)
(12)		CDH74	2621	0	0	0	12	207	615	996	711	80	0	0	0	(12)
(13)		CDH80	720	0	0	0	1	27	163	284	234	11	0	0	0	(13)
(14)	Wind	WSAvg	8.7	9.9	9.9	10.5	9.2	8.9	7.3	6.3	6.8	7.4	9.3	9.7	9.8	(14)
(15)	Precipitation	PrecAvg	20.3	1.0	0.9	1.0	1.2	1.8	2.8	2.3	2.0	1.9	2.1	1.7	1.3	(15)
(16)		PrecMax	30.6	2.3	1.6	2.0	2.7	6.5	7.0	5.8	4.2	4.4	4.7	3.6	2.4	(16)
(17)		PrecMin	11.9	0.1	0.0	0.2	0.0	0.2	0.2	0.0	0.7	0.3	0.4	0.2	0.2	(17)
(18)		PrecStd	4.2	0.5	0.4	0.5	0.8	1.4	1.6	1.5	1.0	1.1	1.1	0.9	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.3	33.7	44.6	75.4	83.2	89.0	91.1	91.7	81.1	66.2	48.9	34.8	(19)
(20)			MCWB	31.9	32.0	38.3	55.1	61.1	65.8	67.3	66.2	58.7	51.7	46.4	33.1	(20)
(21)		2%	DB	31.5	32.2	39.3	68.9	79.7	84.6	86.2	85.9	75.5	61.6	43.4	33.1	(21)
(22)			MCWB	30.6	31.2	34.5	52.7	59.2	65.2	67.9	65.7	58.0	50.0	41.6	31.8	(22)
(23)		5%	DB	29.5	30.3	36.1	64.0	75.7	80.7	83.0	81.7	70.9	56.8	40.1	31.6	(23)
(24)			MCWB	28.6	29.1	33.1	50.0	58.2	63.8	67.0	64.9	56.8	48.1	38.2	30.6	(24)
(25)		10%	DB	26.7	27.6	33.9	58.4	71.4	77.1	79.8	77.2	66.3	52.4	37.1	29.7	(25)
(26)			MCWB	25.7	26.4	31.7	47.3	55.9	62.9	65.7	62.9	55.0	46.0	35.5	28.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.4	32.4	40.1	56.8	64.4	69.8	71.5	69.4	62.8	55.1	46.8	33.7	(27)
(28)			MCD B	33.0	33.0	44.5	72.5	77.4	82.1	84.2	82.3	74.3	60.7	48.6	33.9	(28)
(29)		2%	WB	30.7	31.3	35.3	53.8	61.2	67.8	69.9	67.5	60.3	51.8	42.0	32.2	(29)
(30)			MCD B	31.3	32.0	38.0	65.7	75.3	80.5	81.7	81.4	70.3	58.7	43.2	32.9	(30)
(31)		5%	WB	28.6	29.2	33.5	50.9	59.3	65.9	68.4	65.9	58.2	48.8	38.5	30.6	(31)
(32)			MCD B	29.5	30.3	35.6	62.9	72.6	76.9	79.9	78.6	66.9	55.1	39.7	31.5	(32)
(33)		10%	WB	25.7	26.4	32.0	47.7	57.1	64.1	66.8	64.1	56.2	46.6	35.8	28.7	(33)
(34)			MCD B	26.7	27.5	33.8	57.6	69.6	74.3	77.6	75.3	64.3	51.9	36.9	29.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.6	10.5	11.3	14.5	18.1	17.5	17.6	16.8	14.8	10.5	7.9	9.1	(35)
(36)			MCDBR	9.4	9.7	10.7	21.4	22.4	22.5	21.8	22.3	20.7	17.5	9.0	8.1	(36)
(37)			MCWBR	9.4	9.3	8.0	10.6	9.8	10.0	9.3	8.7	9.8	9.9	8.1	8.0	(37)
(38)		5% WB	MCD BR	9.5	9.2	9.1	21.1	20.9	19.5	19.8	20.2	18.5	15.3	8.4	7.9	(38)
(39)			MCWBR	9.5	9.2	7.5	11.5	10.6	9.9	9.3	9.0	10.1	9.8	8.1	8.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.286	0.310	0.358	0.404	0.389	0.385	0.398	0.421	0.371	0.344	0.310	0.282	(40)	
(41)		taud	2.063	2.056	1.988	2.112	2.298	2.355	2.317	2.234	2.350	2.321	2.205	2.041	(41)	
(42)		Ebn at Noon	197	237	251	258	271	273	267	251	250	227	190	167	(42)	
(43)		Edh at Noon	22	31	43	43	38	36	37	38	30	24	19	18	(43)	
(44)	All-Sky Solar Radiation	RadAvg	270	604	1053	1396	1722	1868	1847	1462	940	440	189	159	(44)	
(45)		RadStd	35	56	113	139	166	203	167	133	123	103	40	40	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) +243	(m) N/A	(46)
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page